BST Construction from Given Elements

Time Complexity

• BST Construction: $O(n^2)$ in the worst case.

• Inorder Traversal: O(n).

• Preorder Traversal: O(n).

• Postorder Traversal: O(n).

Space Complexity

• BST Storage: O(n).

• Recursive Traversals: O(h), where h is the height of the BST.

• Overall Auxiliary Space: O(h).

Here, n is the number of nodes and h is the height of the BST.